

Bio 1A - Lecture 7: Cellular Respiration

Cellular respiration is how cells convert glucose into usable energy.

It has three stages: glycolysis, the citric acid cycle, and the electron transport chain.

Glycolysis takes place in the cytosol and splits glucose into two molecules of pyruvate, producing a small amount of ATP and NADH.

The electron transport chain sits in the inner mitochondrial membrane.

Oxygen is the terminal electron acceptor of the electron transport chain. Without oxygen to accept electrons at the end of the chain, the whole chain backs up and ATP production by oxidative phosphorylation stops. This is why aerobic organisms must breathe.

The proton gradient built across the membrane drives ATP synthase.